SECTIONS OF A SHEAF ON A PROPER SCHEME OVER A HENSELIAN PAIR AGREE WITH SECTIONS ON THE CLOSED FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the topological criterion, the reduction to irreducible X and to the Stein factorisation, the connectedness of the intersection of the closed image with V(I), and the final connected-fibres argument.

Editorial note. The label lemma-zariski-h0-proper-over-henselian-pair names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the six cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The topological h^0 criterion via connectedness of the intersections. Same chapter, the lemma reducing the comparison of sections over X and Z to the connectedness of Y cap Z for every irreducible closed subscheme Y of X. Supplies the criterion that the proof verifies.
- Spectrality of the underlying space of a proper scheme over a henselian pair. Properties, the lemma making the underlying topological space of X spectral. Supplies the first hypothesis of the topological criterion.
- Stein factorisation and integrality over a henselian pair. More on Morphisms, the Stein factorisation theorem, and More on Algebra, the lemmas that an integral extension of a henselian pair is again henselian and that the image of an irreducible scheme meets V(I) in a connected set. Supplies the reduction to geometrically connected fibres and the connectedness of T cap V(I).

Source reference. Original reference 09ZG: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15099–15099. Prerequisite pointer invoked at proof line(s) 18176; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 09XK: sources/stacks-chow/tags.txt, source lines 12533–12533. Prerequisite pointer invoked at proof line(s) 18187; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09Y6: sources/stacks-chow/tags.txt, source lines 12554–12554. Prerequisite pointer invoked at proof line(s) 18192; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original theorem 03H2: sources/stacks-chow/tags.txt, source lines 4549–4549. Prerequisite pointer invoked at proof line(s) 18185; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 094L: sources/stacks-chow/tags.txt, source lines 11554–11554. Prerequisite pointer invoked at proof line(s) 18178; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0377: sources/stacks-chow/tags.txt, source lines 4190–4190. Prerequisite pointer invoked at proof line(s) 18196; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let (A, I) be a henselian pair. Let $f : X \to \mathrm{Spec}(A)$ be a proper morphism of schemes. Let $Z = X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I)$. For any sheaf $\mathcal{F}$ on the topological space associated to X we have $\Gamma(X, \mathcal{F}) = \Gamma(Z, \mathcal{F}|_Z)$.*

Proof. We will use Lemma 09ZG to prove this. First observe that the underlying topological space of X is spectral by Properties, Lemma 094L. Let $Y \subset X$ be an irreducible closed subscheme. To finish the proof we show that $Y \cap Z = Y \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I)$ is connected. Replacing X by Y we may assume that X is irreducible and we have to show that Z is connected. Let $X \to \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the Stein factorization of f (More on Morphisms, Theorem 03H2). Then $A \to B$ is integral and (B, IB) is a henselian pair (More on Algebra, Lemma 09XK). Thus we may assume the fibres of $X \to \mathrm{Spec}(A)$ are geometrically connected. On the other hand, the image $T \subset \mathrm{Spec}(A)$ of f is irreducible and closed as X is proper over A. Hence $T \cap V(I)$ is connected by More on Algebra, Lemma 09Y6. Now $Y \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/I) \to T \cap V(I)$ is a surjective closed map with connected fibres. The result now follows from Topology, Lemma 0377. □